Eunwoo Kim

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RESEARCH INTERESTS

Cognitive Modeling *(Reinforcement Learning, Sequential Sampling Models)*, Joint Modeling of Choice and Response Time, Model-Based Cognitive Neuroscience, Interpretable Machine Learning

EDUCATION

M.S. in Psychology (Quantitative Psychology), Yonsei University Aug. 2026

- Advisor: Inhan Kang · GPA 4.27 / 4.30
- Thesis: *Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior*

B.A. in Psychology; B.S. in Computer Science and Engineering, Seoul National University Aug. 2024

- **Summa Cum Laude**, Department of Psychology · GPA 4.06 / 4.30
- Psychology Thesis: *Causal Inference on COVID-19-Related Events and Well-Being Using Causal Forests*
- Computer Science Thesis: *Application of Basic Prompting Techniques in Korean* (departmental poster presentation)

MANUSCRIPTS

Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior

Eunwoo Kim & Inhan Kang. Submitted to *Psychonomic Bulletin & Review*. [Preprint] · [OSF] *Under review*
Master's Thesis

Reinforcement learning models of the Iowa Gambling Task use choice data alone, and the recovery of their individual-level parameters has received limited scrutiny. This project adds an evidence-accumulation stage so that choices and response times are accounted for by a single mechanism, and examines how well the resulting individual-level parameters are recovered.

- **Model.** Replaced the softmax choice rule of an existing reinforcement learning model for the task with a racing diffusion process, so that trial-by-trial value updating drives the within-trial evidence accumulation that produces each response and response time. The racing formulation extends directly to the four-deck structure of the task.
- **Estimation.** Implemented the model in Stan, writing custom likelihood functions for the joint density of choice and response time and estimating seven individual-level parameters per participant under weakly informative priors using Hamiltonian Monte Carlo.
- **Validation.** Recovered all seven individual-level parameters accurately in a simulation study. Reproduced card-wise choice trajectories, win-stay/lose-shift patterns, and response time distributions in posterior predictive checks. Compared model fit against the prospect valence learning–delta and outcome-representation learning models, with the joint model performing as well or better.
- **Role.** First-authored the manuscript and presented the work at three conferences, including an oral presentation at the International Meeting of the Psychometric Society.

Predicting Fertility Desires Among Premarital Couples: An Interpretable Machine Learning Approach Using Dyadic Data

Eunwoo Kim*, Hyoungbin Moon*, Young-Hoon Kim, & Inhan Kang. (*equal contribution) *In preparation*

Fertility desires within a couple reflect characteristics of both partners. This project predicts them from dyadic survey data using machine learning, and then examines which variables drive the predictions, including the contribution of each partner's characteristics.

- **Model.** Fit random forest models with cross-validation to predict fertility desires from dyadic survey data on premarital couples, using both actor-reported and partner-reported predictors from each couple.
- **Interpretation.** Applied Shapley additive explanation (SHAP) values to quantify how much each variable contributed to the predictions, identifying which characteristics mattered most and how much the partner's characteristics contributed.

– **Role.** Built the full analysis pipeline in Python as co-first author, from preparation of the dyadic data structure through to the interpretation step.

ONGOING PROJECTS

Joint Modeling of Single-Trial Behavior and fMRI Activity: A Factor-Analytic Neural Diffusion Decision Model Applied to Rewarded Word Recognition

Eunwoo Kim*, Seoyoung Choi*, Sungwoo An, Sanghoon Han, & Inhan Kang. (*equal contribution)

Model-based cognitive neuroscience usually estimates cognitive parameters first and relates them to brain activity afterwards, which discards parameter uncertainty and prevents the neural data from informing the cognitive model. This project estimates both together for a rewarded recognition memory task.

– **Model.** Extended the lasso factor-analytic neural diffusion decision model (Kang et al., 2022) to a rewarded word recognition task in which reward is given either for hits or for correct rejections while fMRI is recorded. The brain–behavior link is specified as a factor analysis: single-trial decision parameters are the latent factors, region-of-interest responses are the indicators loading on them, and a Bayesian lasso prior on the loading matrix selects which regions track which component of the decision process.

– **Joint estimation.** Estimated the behavioral and neural components simultaneously rather than in the conventional two-stage sequence, so that uncertainty in the decision parameters carries through to the loadings and the neural data in turn help identify the single-trial parameters.

– **Role.** Responsible for the modeling stage: reformulated the model specification to match the task design and implemented the estimation routine, working from neural data preprocessed by collaborators.

– **Status.** Individual-level fitting complete; currently improving the computational efficiency and numerical stability of the estimation routine, with group-level analysis in progress.

CONFERENCE PRESENTATIONS

Kim, E., & Kang, I. Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior.

International Meeting of the Psychometric Society (IMPS), *Oral presentation* Jul. 2026
Korean Society of Psychological Measurement and Assessment, *Oral presentation* Oct. 2025
Annual Conference of the Korean Psychological Association, *Poster presentation* Aug. 2025

Kim, E., Choi, S., An, S., Han, S., & Kang, I. Joint Modeling of Single-Trial Behavior and fMRI Activity: A Factor-Analytic Neural Diffusion Decision Model Applied to Rewarded Word Recognition.

Korean Society for Computational Neuroscience (CBrain), *Poster presentation* Sep. 2026

HONORS AND AWARDS

Honorable Mention, Outstanding Master's Thesis Award, Department of Psychology, Yonsei University 2026
Future Psychologist Award, Korean Psychological Association 2025
Summa Cum Laude, Department of Psychology, Seoul National University 2024

FELLOWSHIPS AND GRANTS

Graduate Idea Incubating Fellowship (IIF), Yonsei University ($1,500) 2026
Graduate Student Research Assistantship Scholarship, Yonsei University 2025–2026
Teaching Assistant Scholarship, Yonsei University 2024–2025

RESEARCH EXPERIENCE

Lab Manager · Psychometric & Computational Modeling Laboratory Aug. 2025–Aug. 2026
Yonsei University · PI: Inhan Kang

– **Programmed** a PsychoPy survey task for response time measurement in which mouse-to-response distance was held constant across items, so that observed response time differences reflect cognitive rather than motor variation.

– **Managed** data collection from recruitment through completion, running experimental sessions and maintaining participant compensation records.

– **Administered** the laboratory's research operations, including budget and equipment management, participant-payment documentation, and coordination of conference travel for lab members.

Research Assistant · Connectome Laboratory Feb.–Aug. 2024
Seoul National University · PI: Jiook Cha

- **Led** an independent project, which became my undergraduate thesis in psychology, applying causal forests to estimate heterogeneous treatment effects of COVID-19-related stressors on well-being and to identify for whom those effects were strongest rather than estimating an average effect alone.
- **Implemented** the causal forest analysis in R (*grf* package), including estimation of conditional average treatment effects and examination of the respondent characteristics that moderated them.
- **Collaborated** with the Center for Happiness Studies at Seoul National University, which provided the dataset, and jointly interpreted the results with the research team.

Research Assistant · Quantitative Research Collaboratory Jan.–Dec. 2023
University of California, Los Angeles · PI: Amanda K. Montoya

- **Served** as one of several analysts in a lab-internal many-analysts exercise, independently making and documenting analytic decisions on a shared dataset so that variability across analysts could be compared.
- **Conducted** literature reviews for three projects: tree-based machine learning methods in psychological research, SPSS/SAS macros for mediation and moderation analysis, and a registered-reports survey study.
- **Assisted** with survey administration for the registered-reports project.

Research Assistant · Computational Clinical Science Laboratory Aug.–Dec. 2022
Seoul National University · PI: Woo-Young Ahn

- **Wrote** R code visualizing individual differences in performance for a project analyzing data from an orthogonalized go/no-go task.
- **Participated** in a weekly literature group on computational approaches to decision-making and fMRI analysis

COMPUTATIONAL AND TECHNICAL SKILLS

Cognitive Modeling

- Reinforcement learning models of the Iowa Gambling Task (delta-rule value updating, outcome-representation learning, prospect valence learning); sequential sampling models (diffusion decision model, racing diffusion model, linear ballistic accumulator model); joint reinforcement learning–sequential sampling models; neural diffusion decision models with factor-analytic structure.
- *Applied in*: the Iowa Gambling Task reinforcement learning–sequential sampling model; the lasso factor-analytic neural diffusion decision model.

Bayesian Estimation — Stan / RStan

- **Model implementation**: custom likelihood functions for the joint density of choice and response time, with the comparison models implemented in the same framework.
- **Estimation and diagnostics**: weakly informative and shrinkage (Bayesian lasso) priors, convergence diagnostics (R-hat, effective sample size, divergent transitions), posterior predictive checking, and simulation-based parameter recovery.
- *Applied in*: the Iowa Gambling Task reinforcement learning–sequential sampling model; the lasso factor-analytic neural diffusion decision model.

Machine Learning and Causal Inference — Python, R

- **Interpretable machine learning in Python**: random forest models with cross-validation, and Shapley additive explanation (SHAP) values for variable-level attribution.
- **Causal inference in R**: causal forests (*grf* package) for heterogeneous treatment effect estimation, conditional average treatment effects, and identification of moderating characteristics.
- *Applied in*: the fertility desires prediction project; the COVID-19 stressor heterogeneous treatment effect project.

Experiment Programming and Data Collection

- **PsychoPy**: design and implementation of response-time-sensitive tasks in Python, including control of motor confounds across items; deployment in live data collection.
- *Applied in*: Yonsei University lab manager role (task programming and participant sessions).

Neuroimaging and Structural Equation Modeling Software — coursework level

- **SPM (MATLAB)**: fMRI preprocessing pipeline practiced in graduate neuroimaging coursework.

– **Mplus**: structural equation modeling syntax covered in graduate coursework.

Programming Languages

– **R** (analysis and Bayesian model fitting through RStan), **Stan** (probabilistic modeling), **Python** (machine learning, experiment programming), **MATLAB**.

– Formal training in computer science (B.S., Seoul National University) supports the algorithmic implementation and computational efficiency of my modeling work.

TEACHING EXPERIENCE

Teaching Assistant, Department of Psychology, Yonsei University

Graded assignments and provided individualized written feedback that identified the source of conceptual and coding errors and demonstrated the appropriate approach.

Bayesian Statistics for Psychology (Graduate) Fall 2025

Psychological Statistics (Undergraduate) Spring 2025

Data Science for Analysis of Human Behavior and Psychology (Undergraduate) Fall 2024

SELECTED COURSEWORK

Statistics and Methods — Mathematical Statistics, Linear Models, Generalized Mixed Models for Data Science, Structural Equation Modeling, Machine Learning Theory and Application, Linear Algebra, Psychometrics

Psychology and Neuroscience — Seminar in Quantitative Psychology, Seminar in Experimental Psychology, Advanced Cognitive Science, Neuroimaging, Neuroscience

REFERENCES

Inhan Kang, Ph.D. — Assistant Professor, Department of Psychology, Yonsei University
M.S. thesis advisor; supervises my work on the reinforcement learning–sequential sampling model and on the joint neural–behavioral modeling project.
qpsy@yonsei.ac.kr | [homepage]

Sanghoon Han, Ph.D. — Professor, Department of Psychology, Yonsei University
M.S. thesis committee member; collaborator on the joint neural–behavioral modeling project, which analyzes data collected in his laboratory.
sanghoon.han@yonsei.ac.kr | [homepage]

Hanjoe Kim, Ph.D. — Associate Professor, Department of Psychology, Yonsei University
M.S. thesis committee member; instructor for my graduate coursework in quantitative methods and a continuing source of methodological guidance.
hanjoekim@yonsei.ac.kr | [homepage]

Jiook Cha, Ph.D. — Associate Professor, Department of Psychology, Seoul National University
Supervised my independent project on heterogeneous treatment effects of COVID-19-related stressors during my undergraduate research assistantship in the Connectome Laboratory.
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